```
# 1. Print all even numbers from 1 to 50 without arguments

def even_numbers():

    for i in range(1, 51):

        if i % 2 == 0:

            print(i)



even_numbers()




# 2. Sum of two numbers using arguments

def add(a, b):

    print("Sum =", a + b)



add(10, 20)




# 3. Return sum, average, minimum and maximum from a list

def list_details(lst):

    s = sum(lst)

    avg = s / len(lst)

    return s, avg, min(lst), max(lst)
```

```
print(list_details([10, 20, 30, 40, 50]))



# 4. Largest of three numbers using return

def largest(a, b, c):

    return max(a, b, c)



print("Largest =", largest(10, 50, 30))



# 5. Positional and keyword arguments together

def student(name, age, course):

    print("Name:", name)

    print("Age:", age)

    print("Course:", course)



student("Harini", age=20, course="Python")



# 6. Area of rectangle using positional arguments

def area(length, breadth):
```

```
    print("Area =", length * breadth)


area(10, 5)



# 7. Simple Interest with default rate 5%

def simple_interest(p, t, r=5):

    si = (p * t * r) / 100

    print("Simple Interest =", si)


simple_interest(1000, 2)



# 8. Power of a number with default exponent 2

def power(num, exp=2):

    print("Result =", num ** exp)


power(5)

power(5, 3)



# 9. Sum using *args
```

```
def total(*args):
    print("Sum =", sum(args))

total(10, 20, 30, 40)


# 10. Largest value using *args
def largest_value(*args):
    print("Largest =", max(args))

largest_value(10, 50, 30, 80, 20)


# 11. Display all key-value pairs using **kwargs
def display(**kwargs):
    for key, value in kwargs.items():
        print(key, ":", value)

display(name="Harini", age=20, city="Salem")


# 12. Display employee details using **kwargs
```

```
def employee(**kwargs):
    for key, value in kwargs.items():
        print(key, ":", value)

employee(id=101, name="Ravi", salary=25000)
```

```
# 13. Lambda function to find square
square = lambda x: x * x
print(square(5))
```

```
# 14. Lambda function to find larger number
large = lambda a, b: a if a > b else b
print(large(10, 20))
```

```
# 15. Convert Celsius to Fahrenheit using map()
celsius = [0, 20, 30, 40]
fahrenheit = list(map(lambda c: (c * 9/5) + 32, celsius))
print(fahrenheit)
```

```
# 16. Find squares using map()

nums = [1, 2, 3, 4, 5]

squares = list(map(lambda x: x**2, nums))

print(squares)




# 17. Extract even numbers using filter()

nums = [1, 2, 3, 4, 5, 6, 7, 8]

even = list(filter(lambda x: x % 2 == 0, nums))

print(even)




# 18. Extract numbers greater than 100 using filter()

nums = [50, 120, 80, 200, 150]

greater = list(filter(lambda x: x > 100, nums))

print(greater)




# 19. Recursive function to calculate power

def power_rec(a, b):

    if b == 0:
```

```
        return 1
    return a * power_rec(a, b - 1)

print(power_rec(2, 3))



# 20. Recursive Fibonacci series up to N terms
def fibonacci(n):
    if n <= 1:
        return n
    return fibonacci(n-1) + fibonacci(n-2)

n = 6
for i in range(n):
    print(fibonacci(i), end=" ")
```